

Balmy Beach Community Day Care (Toronto) ^[6]



160+ kids

- Fast-paced, high-supervision environment
- Field trips require staff to apply wristbands to every child before departure.
- Wristband application creates a significant staff burden.



There is an opportunity to **improve the wristband application process** within the daycare environment.

Design Objectives



Improve Efficiency and Safety

- Reduce time required for wristband application
- Ensure consistent application process



Reduce Staff Burden

- Fewer manual steps
- More focus on supervision



Enhance Child Experience

- Low-anxiety, fun and engaging
- Smooth, stress-free process

Impact

Children

Low-anxiety,
Fast process,
Safe application,
Comfortable
experience

Staff

Reduced workload,
Minimal interaction,
Improved efficiency

Next Steps

Integrate all subsystems into a fully functional prototype

Refine feeding mechanism

Evaluate durability and long-term wear

Select and validate materials for safety and performance

Upgrade to a higher-precision force sensor for adjustable control

Integrate Excess Management System

Work Cited:

- [1]Norman, D. A. (2013). The Design of Everyday Things
- [2]Wickens, C. D., Lee, J. D., Liu, Y., & [3]Becker, S. (2004). An Introduction to Human Factors Engineering
- [4]Kaminski, T. R., et al. (2005)
- [5]Mathiowetz, V., Wiemer, D. M., & Federman, S. M. Grip and pinch strength: norms for 6–19-year-olds.
- [6]Balmy Beach Community Daycare (Toronto). Stakeholder consultation and field observations.
- [7]Lidwell, W., Holden, K., & Butler, J. (2010).

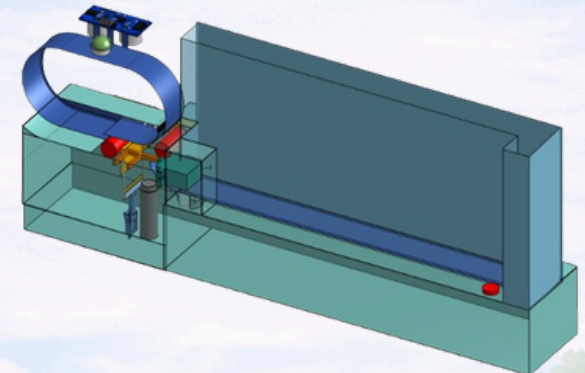
Scan me to see the instructional kids video for SnapBuddy



The SNAP BUDDY

AUTOMATED WRISTBAND APPLICATION SYSTEM

The faster, effortless way to get the field trip started.



Praxis II Team 11

Bice Vercelli
Katie Di Gregorio
Jincheng Luo
Harshan Baranthagan

Step by Step

1. Put your hand inside the wrapping rail loop (1). The ultrasonic sensor recognizes this and signals the motors to turn respectively.
2. A yellow light will turn on to indicate to keep your hand in place.
3. The bracelet is then fed through roller 1 (6) and loops through the wrapping rail (3).
4. Motor 2 (7) tightens the wristband by pushing the bracelet towards the buttoning/cutting mechanism (4).
5. Once the cutting and buttoning mechanism is complete, the bracelet is fully applied to your wrist.
6. Finally, the green light turns on. Gently remove your hand.

ANY EMERGENCIES: PRESS RED BUTTON.

Key Requiriments

Child Engagement (R1) [4]

- Visual & sensory feedback • Play-based engagement

Application Time (R2) [3]

- < 22.7 s application (post-hand placement) • Continuous, streamlined process

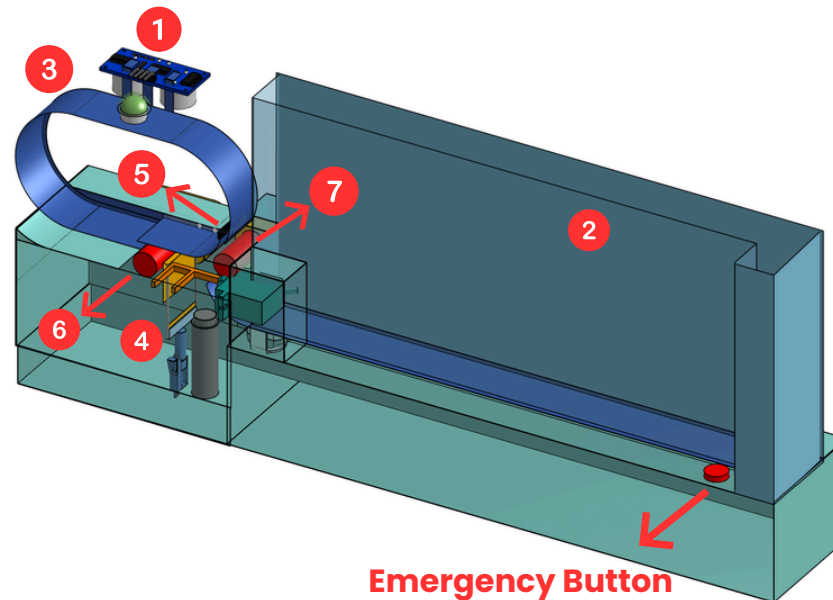
Force Safety (R3) [5]

- 3–5 N controlled force • < 10 kPa contact pressure • Safe for child interaction

Staff Burden (R4) [7]

- ≤ 6 manual actions Low-touch operation • Minimal cognitive load

The SNAP BUDDY



Design Components

1. Ultrasonic Sensor
2. Bracelet Storage / Feeding
3. Wrapping Rail
4. Buttoning/Cutting Mechanisms
5. Force Sensitive Resistor
- 6 & 7. DC Motors & Friction Rollers
 - 6 → motor 1
 - 7 → motor 2



Verification & Testing

Child Engagement (R1)

- Tested with children at Balmy Beach Daycare
- Positive responses to colors, lights, and instructional video
- No visible signs of anxiety during interaction

Application Time (R2)

- Manual baseline: 22.7 s (15 trials)
- System time calculated using motor speed + path length
→ ≈ 10.3 s ≈ 2× faster than current process

Force Safety (R3)

- Measured using thin-film pressure sensor
- Recorded pressure ≈ 10 kPa (within safe limits)

Wrist Size Fit (R7)

- Tested on users with varying hand sizes
- Child with largest wrist at daycare successfully fit → confirms upper-range accommodation